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Time taken	41 mins 19 secs
Marks	52.70/60.00
Grade	27.67 out of 31.50 (87.83%)

Question 1

Complete

Mark 0.00 out of 1.00

Which of the following statements is true about weak ETags compared to strong ETags?

- ☐ a. Weak ETags are only valid for resources under 1 KB.
- ☒ b. Weak ETags require mandatory use of the `If-Modified-Since` header.
- ☐ c. Weak ETags ensure byte-for-byte identical content.
- ☐ d. Weak ETags are case-sensitive, while strong ETags are not.
- ☐ e. Weak ETags can be used when content differences are insignificant.

Question 2

Complete

Mark 1.00 out of 1.00

Which statements about HATEOAS are correct?

- ☐ a. Clients must know all valid resource URLs in advance
- ☒ b. HATEOAS ensures clients discover actions dynamically through responses
- ☐ c. The server sends a finite state machine definition to the client
- ☐ d. HATEOAS is optional for RESTful systems
- ☒ e. The server provides hyperlinks to the next possible states for the client

Question 3

Complete

Mark 0.00 out of 1.00

What does HATEOAS stand for?

Answer:

Question 4

Complete

Mark 1.00 out of 1.00

How does the `If-Match` header in a conditional PUT request improve data consistency?

- ☒ a. It prevents updates unless the current ETag matches the client's provided ETag.
- ☐ b. It bypasses server-side validation checks for efficiency.
- ☐ c. It allows overwriting of the resource regardless of its current state.
- ☐ d. It verifies that the resource exists at a specific URL.
- ☐ e. It ensures that the requested resource is publicly accessible.

Question 5

Complete

Mark 1.00 out of 1.00

What is the main consequence of using hyperlinks in REST responses?

- ☐ a. Clients must predefine all URLs for efficient interaction
- ☐ b. Servers must restrict the number of hyperlinks in responses
- ☒ c. Clients only need to know the dispatcher address to interact with the API
- ☐ d. Clients can perform all operations without server-provided links
- ☐ e. Hyperlinks reduce the server's responsibility for state management

Question 6

Complete

Mark 1.00 out of 1.00

What is the name of the response header that the server uses to inform the client about the new resource's URI?

Answer:

Question 7

Complete

Mark 1.00 out of 1.00

Which of the following is NOT true about gRPC?

- ☐ a. Server stubs can be extended in gRPC.
- ☐ b. gRPC uses Protocol Buffers (Protobuf) for marshalling data.
- ☒ c. Client stubs must be manually written by the developer.
- ☐ d. gRPC supports multiple programming languages.
- ☐ e. gRPC is used by many modern software systems.

Question 8

Complete

Mark 0.00 out of 1.00

Which of the following statements regarding a POST request are correct?

- ☒ a. Server sends a representation of the resource as response body
- ☐ b. Server sends the new URI in the response body
- ☐ c. Server replies with status code 200 when successful
- ☒ d. Server sends the new URI as Location header
- ☒ e. To be used to create a resource on the server
- ☒ f. Server replies with status code 201 when successful

Question 9

Complete

Mark 1.00 out of 1.00

Map the HTTP status code groups to their meaning:

4xx Client request message contained an error

3xx Client is requested to ask somewhere else

2xx Request was processed successfully

5xx Server had a problem to process this request

1xx Informational, request is still processed

Question 10

Complete

Mark 1.00 out of 1.00

What hyperlinks should a server provide in its responses?

- ☐ a. No hyperlinks, as clients should already know all URLs
- ☐ b. Hyperlinks only for critical server actions
- ☐ c. Hyperlinks for only the current resource
- ☒ d. Hyperlinks for every possible next action the client can take
- ☐ e. Hyperlinks for all resources in the API

Question 11

Complete

Mark 1.00 out of 1.00

What is the purpose of a self-link in a REST resource?

- ☐ a. To link the resource to its parent resource
- ☒ b. To provide the URL for accessing the resource itself
- ☐ c. To store metadata about the resource
- ☐ d. To specify the HTTP methods available for the resource
- ☐ e. To redirect clients to a related resource

Question 12

Complete

Mark 1.00 out of 1.00

What is the purpose of media types in HTTP?

- ☐ a. To specify the language of the content
- ☐ b. To encrypt data sent in HTTP requests
- ☐ c. To set the expiration time of the HTTP message
- ☐ d. To identify the client's browser type
- ☒ e. To define the data format of the message body

Question 13

Complete

Mark 0.70 out of 1.00

Why does it make sense that all resources in a collection response contain a self link?

- ☐ a. The response must not have more than one Link header
- ☐ b. It ensures RESTful principles are upheld by making resources discoverable
- ☒ c. It allows clients to interact directly with individual resources without needing to construct URLs manually
- ☐ d. Self links are redundant if the server provides a base URL
- ☒ e. It simplifies hypermedia-driven navigation by providing a consistent reference to each resource

Question 14

Complete

Mark 0.00 out of 1.00

What can be changed in the backend without breaking the client?

- ☒ a. Relation names
- ☐ b. URL to resources
- ☒ c. Database Management System
- ☒ d. Caching Strategy
- ☐ e. Authentication mechanism
- ☐ f. URL of the dispatcher

Question 15

Complete

Mark 1.00 out of 1.00

Which of the following statements about parameter passing in gRPC is correct?

- ☒ a. Call-by-value is used for primitive data types.
- ☐ b. Parameters are passed by URL encoding.
- ☐ c. Call-by-copy is used exclusively.
- ☐ d. Call-by-reference is easy to implement.
- ☐ e. Call-by-pointers is a common method.

Question 16

Complete

Mark 1.00 out of 1.00

What does it mean that HTTP is a stateless protocol?

- ☒ a. Each HTTP request is independent and does not retain user session information
- ☐ b. User authentication is automatically handled by the HTTP protocol
- ☐ c. HTTP maintains a continuous connection with the server for state management
- ☐ d. The HTTP protocol can only be used for stateful communications
- ☐ e. HTTP requests must be made in a sequential order

Question 17

Complete

Mark 1.00 out of 1.00

Is it good practise in REST APIs to provide a URL to login a user, for example /api/login

- ☐ True
- ☒ False

Question 18

Complete

Mark 1.00 out of 1.00

How does the stateless nature of HTTP impact web application development?

- ☐ a. HTTP requests are processed slower due to state management
- ☐ b. Web applications do not need to manage user sessions
- ☒ c. State must be maintained through other means, like cookies or session storage
- ☐ d. Each HTTP request contains the entire history of user-server interaction
- ☐ e. Server resources are heavily used to maintain the state

Question 19

Complete

Mark 1.00 out of 1.00

What happens when a client sends a conditional GET request with an `If-None-Match` header, and the ETag matches the server's current representation of the resource?

- ☐ a. The server responds with a 403 Forbidden status code.
- ☐ b. The server responds with a 404 Not Found status code.
- ☐ c. The server responds with a 200 OK status code and the full resource.
- ☒ d. The server responds with a 304 Not Modified status code.
- ☐ e. The server ignores the ETag and sends the full resource regardless.

Question 20

Complete

Mark 1.00 out of 1.00

An HTTP verb is safe if ...

- ☐ a. the request body can contain sensitive information like passwords
- ☐ b. it can be sent many times and always has the same effect
- ☒ c. it never changes data on the server
- ☐ d. it never deletes data
- ☐ e. it only creates data on the server but never changes data

Question 21

Complete

Mark 1.00 out of 1.00

What happens if the resource name in the URLs changes from "Person" to "User"?

- ☐ a. The client will adjust dynamically and continue functioning
- ☐ b. The change will have no effect on the client
- ☒ c. The client will break and require source code modifications
- ☐ d. The client will cache the old URLs and continue using them
- ☐ e. The server will automatically redirect the client to the new URL

Question 22

Complete

Mark 1.00 out of 1.00

Which of the following statements are correct?

- ☐ a. REST APIs and Web APIs belong to different API paradigms
- ☐ b. REST APIs are just another name for Web APIs
- ☒ c. REST APIs and Web APIs belong the same API paradigms
- ☐ d. REST APIs are a super-set of Web APIs
- ☒ e. REST APIs are a sub-set of Web APIs

Question 23

Complete

Mark 1.00 out of 1.00

Which of the following is a correct way to define a protobuf message field?

- ☐ a. name: string = 1;
- ☐ b. bool name = "true";
- ☐ c. int32 name = "name";
- ☒ d. string name = 1;
- ☐ e. message name = "name";

Question 24

Complete

Mark 1.00 out of 1.00

Which of the following statements about GraphQL is true?

- ☐ a. GraphQL is designed to replace SQL
- ☐ b. GraphQL requires using the REST architecture
- ☒ c. GraphQL allows clients to specify exactly what data they need
- ☐ d. GraphQL is specifically for working with graph databases
- ☐ e. GraphQL does not support nested queries

Question 25

Complete

Mark 1.00 out of 1.00

What is the n+1 problem in the context of GraphQL?

- ☐ a. It is a method for caching queries in GraphQL
- ☐ b. It describes a bug that causes the client to crash after (n+1) requests
- ☐ c. It is a performance optimization technique in GraphQL
- ☐ d. It is an issue where the server sends (n+1) times more data than necessary
- ☒ e. It refers to the situation where a client needs to make (n+1) requests to fetch related data

Question 26

Complete

Mark 0.00 out of 1.00

How does GraphQL address the problem of under-fetching compared to REST?

- ☐ a. By caching responses on the server
- ☐ b. By providing default responses with all possible data
- ☒ c. By requiring clients to send multiple requests for related data
- ☐ d. By allowing clients to request exactly the data they need in a single query
- ☐ e. By using fixed endpoints that return complete data sets

Question 27

Complete

Mark 1.00 out of 1.00

Which of the following statements regarding a DELETE request are correct?

- ☐ a. Server replies with status code 405 if the resource doesn't exist
- ☒ b. Server replies with status code 204 if deleting was successful
- ☒ c. Server replies with status code 410 if the resource was already deleted
- ☒ d. Deleting a resource is dependent on the semantic of the application
- ☐ e. REST APIs don't provide physically deleting a resource, the resource is only marked as deleted

Question 28

Complete

Mark 0.00 out of 1.00

Identify correct aspects of HTTP requests

- ☒ a. Headers can include information like content type and caching
- ☒ b. The HTTP body specifies the target resource
- ☐ c. Status codes are sent in the request
- ☒ d. The HTTP method determines the action on the resource
- ☐ e. The URI specifies the HTTP method

Question 29

Complete

Mark 1.00 out of 1.00

After the client sends a PUT request, the server informs the client that another client updated this resource earlier by sending which status code?

Answer: 409

Question 30

Complete

Mark 1.00 out of 1.00

In a RESTful system, what does the server transmit to clients in response messages?

- ☐ a. The previous state
- ☒ b. All states that are possible next states for the client with regard to the current state
- ☐ c. The dispatcher state
- ☐ d. A set of all states that are visible for this client in this application
- ☐ e. The whole definition of the finite state machine

Question 31

Complete

Mark 1.00 out of 1.00

What is the main advantage of using GraphQL?

- ☐ a. Providing a new type of database query language
- ☐ b. Simplifying backend implementations
- ☐ c. Making it easier to cache requests
- ☒ d. Reducing network requests and bandwidth usage
- ☐ e. Allowing the server to control the data returned

Question 32

Complete

Mark 0.00 out of 1.00

What exactly is the meaning of a 4xx status code?

- ☒ a. Client request contained an error, client may send this request again after some time
- ☐ b. Client request contained an error but the server was able to process it anyhow
- ☐ c. Client request contained an error, client must not send this request again
- ☐ d. Client request was correct but the server was not able to process it

Question 33

Complete

Mark 1.00 out of 1.00

What does the `rel` attribute in an RFC 5988 link represent?

- ☐ a. The authentication method required for the link
- ☐ b. The HTTP method used to access the link
- ☐ c. The absolute URL of the hyperlink
- ☐ d. The type of the data expected by the link
- ☒ e. The relationship type of the hyperlink

Question 34

Complete

Mark 1.00 out of 1.00

Which of the following best describes the difference between GraphQL and REST?

- ☐ a. GraphQL and REST cannot be used together in the same application
- ☐ b. GraphQL is only for real-time data, whereas REST is for static data
- ☐ c. GraphQL is used for backend development, while REST is used for frontend development
- ☒ d. GraphQL allows clients to query exactly what they need, whereas REST typically returns fixed data structures
- ☐ e. GraphQL is more secure than REST

Question 35

Complete

Mark 1.00 out of 1.00

A status code of 303 falls under which category of HTTP status codes?

- ☐ a. Client Error Responses
- ☐ b. Informational Responses
- ☐ c. Successful Responses
- ☒ d. Redirection Messages
- ☐ e. Server Error Response

Question 36

Complete

Mark 1.00 out of 1.00

What is the primary cause of over-fetching in REST APIs?

- ☐ a. The server fails to filter the data properly
- ☒ b. The server provides more data than the client needs
- ☐ c. The client sends multiple requests for the same data
- ☐ d. The server does not have sufficient data to respond
- ☐ e. The client requests data that does not exist

Question 37

Complete

Mark 1.00 out of 1.00

Which of the following statements regarding a PUT request are correct?

- ☒ a. PUT must be used to update a resource
- ☐ b. POST and PUT requests are the same
- ☐ c. PUT request without request body deletes the resource on the server
- ☒ d. PUT must be used to create a resource if the client knows the identifier
- ☒ e. Server replies with status code 200 when update was successful
- ☐ f. Server replies with status code 204 and the new representation when successful

Question 38

Complete

Mark 1.00 out of 1.00

Which HTTP verb is not idempotent?

Answer: POST

Question 39

Complete

Mark 1.00 out of 1.00

Why must hyperlinks be provided for every client request?

- ☐ a. To improve the client's authentication process
- ☒ b. To guide the client towards the next valid action
- ☐ c. To ensure the client stays connected to the server
- ☐ d. To reduce the number of requests made by the client
- ☐ e. To enforce server-side caching

Question 40

Complete

Mark 1.00 out of 1.00

Which of the following is an example of a mutation in GraphQL?

- ☐ a. `query { person(id: 1) { firstName lastName } }`
- ☐ b. `query { create(firstName: "John", lastName: "Doe") { firstName lastName } }`
- ☐ c. `mutation { getPersons { firstName lastName } }`
- ☒ d. `mutation { create(firstName: "John", lastName: "Doe") { firstName lastName } }`
- ☐ e. `mutation { persons { firstName lastName } }`

Question 41

Complete

Mark 1.00 out of 1.00

Web or HTTP-based APIs are examples for what kind of API?

- ☒ a. Document-oriented API
- ☐ b. Data-oriented API
- ☐ c. Function-oriented API
- ☐ d. Object-oriented API

Question 42

Complete

Mark 1.00 out of 1.00

What is a good "next state" after the client has deleted a resource?

- ☐ a. The confirmation dialog "Are you sure?"
- ☒ b. The collection of all resources of this type
- ☐ c. The dispatcher state
- ☐ d. The resource itself
- ☐ e. The state to undelete the resource

Question 43

Complete

Mark 1.00 out of 1.00

Which of the following statements regarding state-less-ness are correct?

- ☒ a. The client has to send all information necessary for the server to process the request.
- ☐ b. State-less-ness is practically irrelevant - nobody cares about it
- ☒ c. State-less-ness is one prerequisite for scalable Web servers
- ☐ d. State-less-ness is a nice-to-have feature but sometimes too difficult to realise and, therefore, optional
- ☒ e. The server does not know that two requests of the same client are related.

Question 44

Complete

Mark 1.00 out of 1.00

Mark all valid URIs

- ☐ a. http://example.org:8080\path\to\document?parameter=hello
- ☐ b. http://example.org?parameter=1?query=
- ☐ c. http://example.org:-1?parameter=1&query=hello
- ☐ d. https://example.org:443\&a=1&query=hello
- ☒ e. http://example.org:80?parameter=1&query=Hello
- ☒ f. http://example.org/path/to/document.html

Question 45

Complete

Mark 1.00 out of 1.00

In a RESTful system, the client must know ...

- ☐ a. only the structure of all URIs as defined by the backend developers
- ☒ b. only the URI of the dispatcher state
- ☐ c. only the common prefix of all URIs
- ☐ d. only the name of the functions, e.g. "createPerson" or "getLocations"
- ☐ e. only the names of the resources, e.g. "persons" or "locations"

Question 46

Complete

Mark 1.00 out of 1.00

What is a common use case for GraphQL's fragment feature?

- ☒ a. Reusing sets of fields across multiple queries
- ☐ b. Caching query results
- ☐ c. Handling errors in queries
- ☐ d. Performing batch operations
- ☐ e. Defining input types for mutations

Question 47

Complete

Mark 1.00 out of 1.00

In order to create a new resource without identifier, a client must

- ☐ a. send a GET request with the attributes of the resource as query parameters
- ☐ b. send a POST request with id and with the representation of the resource as request body
- ☐ c. send a POST request with id and with the representation of the resource in the URL
- ☒ d. send a POST request without id and with the representation of the resource as request body
- ☐ e. send a PUT request with the representation of the resource as request body

Question 48

Complete

Mark 1.00 out of 1.00

Which HTTP method's success typically does not involve specifying a media type in the response?

- ☐ a. POST
- ☐ b. GET
- ☐ c. PUT
- ☒ d. DELETE
- ☐ e. PATCH

Question 49

Complete

Mark 1.00 out of 1.00

What is the consequence of changing a field's type in a gRPC message for backward compatibility?

- ☐ a. It leads to warnings during Protocol Buffers compilation but not runtime errors.
- ☐ b. The server automatically adapts to the new type.
- ☒ c. It can cause serialization/deserialization errors in older clients.
- ☐ d. It has no impact on compatibility as long as the field is used correctly.
- ☐ e. It requires all clients to be updated immediately.

Question 50

Complete

Mark 1.00 out of 1.00

Select all correct statements about HTTP being stateless

- ☒ a. State can be maintained using mechanisms like cookies or tokens
- ☒ b. Each HTTP request is independent and does not retain user session information
- ☐ c. The HTTP protocol can only be used for stateful communications
- ☐ d. HTTP maintains a continuous connection with the server for state management
- ☐ e. HTTP requests must be made in a sequential order

Question 51

Complete

Mark 1.00 out of 1.00

What is one benefit of using GraphQL over traditional REST APIs?

- ☒ a. Reduced over-fetching and under-fetching of data
- ☐ b. Built-in data caching
- ☐ c. Automatic scalability
- ☐ d. Less setup and configuration
- ☐ e. Simpler authentication

Question 52

Complete

Mark 1.00 out of 1.00

What does the 5 URLs pattern represent in a RESTful service?

- ☐ a. An approach to map URLs directly to backend methods
- ☐ b. A naming convention for database tables
- ☐ c. A method to ensure high availability in RESTful services
- ☐ d. A design pattern for building monolithic applications
- ☒ e. A set of standard URLs for CRUD operations on a resource

Question 53

Complete

Mark 1.00 out of 1.00

What is the main advantage of clients knowing only one URL?

- ☒ a. It decouples client logic from server-side URL changes
- ☐ b. It minimizes the number of API endpoints needed
- ☐ c. It reduces server response times
- ☐ d. It simplifies the server's API documentation
- ☐ e. It increases the client's request efficiency

Question 54

Complete

Mark 1.00 out of 1.00

The basic principle of hypertext is

- ☐ a. to have a version history of a document
- ☒ b. to have references (links) between documents, which results in a graph of documents
- ☐ c. to be written in a markup language like HTML or XML

Question 55

Complete

Mark 1.00 out of 1.00

Why are absolute URLs recommended for hyperlinks in the response header?

- ☐ a. They simplify server-side link generation
- ☐ b. They prevent URL duplication in responses
- ☐ c. They are mandatory for secure communications
- ☐ d. They are required by all HTTP standards
- ☒ e. They ensure the client can resolve the link without context

Question 56

Complete

Mark 1.00 out of 1.00

Map the following HTTP verb to their meaning:

GET	Clients request a resource from the server
DELETE	Clients delete an existing resource on the server
POST	Clients send a resource to the server
PUT	Clients update an existing resource on the server

Question 57

Complete

Mark 1.00 out of 1.00

What is the dispatcher address in a REST API?

- ☐ a. A temporary URL for server maintenance purposes
- ☐ b. An optional URL used for testing server status
- ☐ c. The default page shown when no specific resource is requested
- ☐ d. A reserved endpoint for client authentication
- ☒ e. A single starting URL that provides access to other resources via hyperlinks

Question 58

Complete

Mark 1.00 out of 1.00

What is the primary role of an ETag in HTTP caching?

- ☐ a. It sets the maximum cache duration for the resource.
- ☐ b. It specifies the URL of the resource.
- ☒ c. It uniquely identifies a specific version of the resource.
- ☐ d. It provides the last modified timestamp of a resource.
- ☐ e. It determines the content type of the resource.

Question 59

Complete

Mark 1.00 out of 1.00

How does gRPC ensure backward compatibility in protobuf messages?

- ☐ a. By using JSON format
- ☐ b. By relying on schema-less data exchange
- ☒ c. By ensuring field numbers (tags) remain unique and consistent
- ☐ d. By using versioned APIs
- ☐ e. By converting messages to XML

Question 60

Complete

Mark 1.00 out of 1.00

Which of the following statements regarding a GET request are correct?

- ☒ a. To be used to request a resource representation
- ☐ b. Server replies with status code 500 when requested media type is not available
- ☐ c. Server replies with status code 204 when successful
- ☒ d. Server replies with status code 404 when resource not found
- ☒ e. Server replies with status code 200 when successful